Unit - 4
Hydrogen

EVALUATION

I Choose the best answer:

1. (c)
2. (c) $CO + H_2$
3. (b)
4. (d) group one elements
5. (c) 1p+2n
6. (a) Palladium, Vanadium
7. (a)
8. (a) 1.2 g
9. (d) EDTA
10. (c) $CaCl_2$
11. (a) sodium aluminium silicate
12. (a)
13. (c) $CrO(O_2)_2$
14. (c) 5/2
15. (d) 8.4
16. (d) sp^3 and sp^3
17. (c) monobasic acid
18. (a) tetrahedrally by 4-H atoms
19. (b) intra-molecular hydrogen bonding and inter molecular hydrogen bonding
20. (c) both (a) & (b)
21) (c) amphoteric oxide

Key for multiple choice questions:

1) Option (c)

Correct statement : Hydrogen has three isotopes of which **protium** is the most common.

2) Option (c)

$CO + H_2$ - Water gas

3) Option (b)

Correct statement : Ortho isomer - one nuclear spin

Para isomer - zero nuclear spin

4) Option (d)

eg : Sodium hydride ($Na^+ H^-$)

5) Option (c)

$_1T^3$ ($1e^-$, 1p, 2n)

7) Option (a)

$Ca^{2+} + Na_2CO_3 \rightarrow CaCO_3 \downarrow + 2Na^+$

8) mass of deuterium = 2 × mass of protium

If all the 1.2 g hydrogen is replaced with deuterium, the weight will become 2.4g. Hence the increase in body weight is (2.4 – 1.2 = 1.2 g)

Option (a)

9) EDTA (option (d))

10) Permanent hardness if water is due to the presence of the chlorides, nitrates and sulphates of Ca^{2+} and Mg^{2+} ions.

Option (c) $CaCl_2$

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